UL Solutions

An introduction to UL Solutions and HOMER® software



Safety. Science. Transformation.™

© 2022 UL LLC. All rights reserved.

The future is hybrid

Solar

Storage

Microgrids

DER

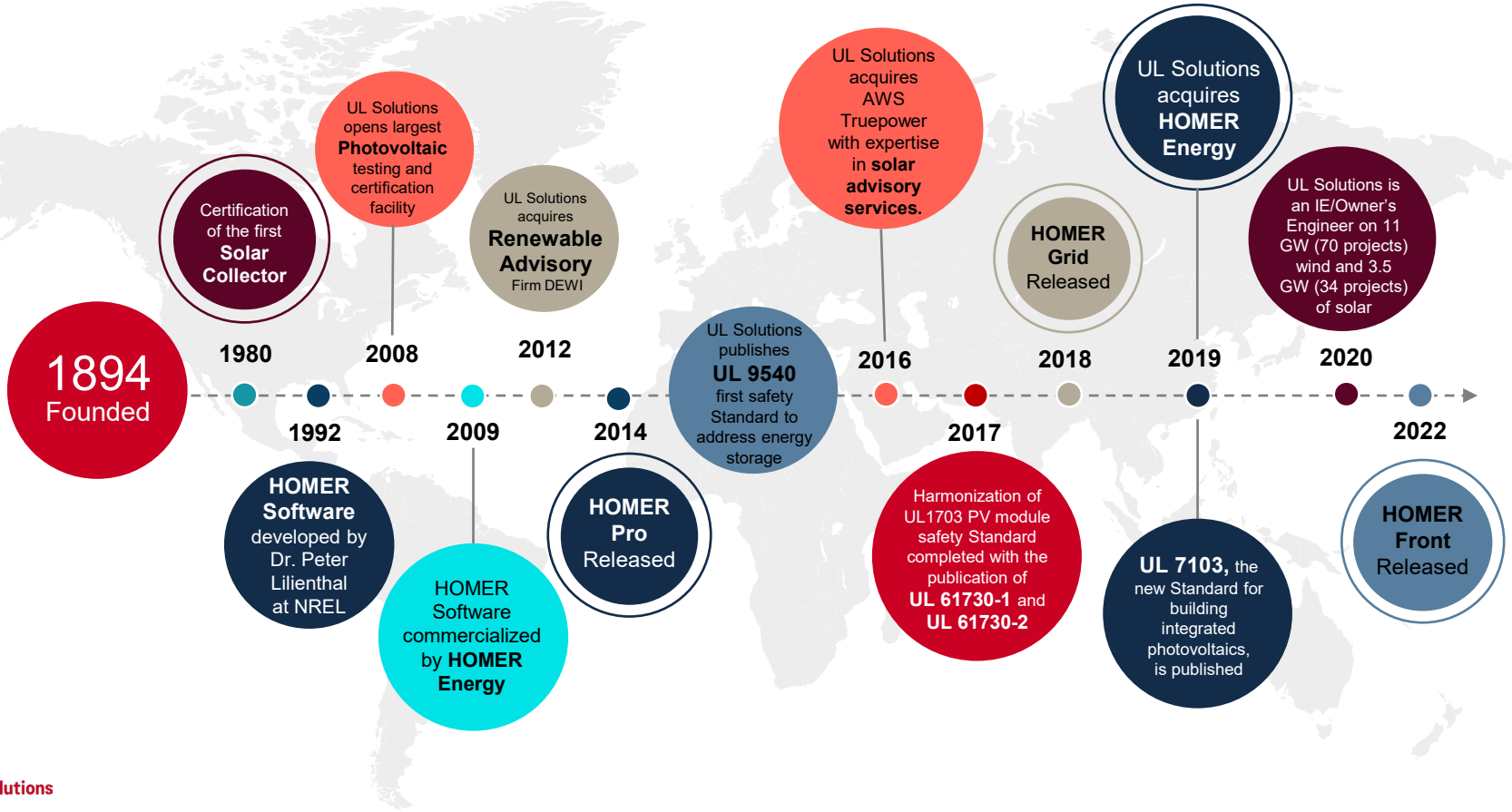
Utility-scale



ASP22CS460808

UL Solutions

UL Solutions: A history of trust



Key office locations - renewables

500+
RENEWABLE ENERGY EXPERTS

35+ years of EXPERIENCE IN RENEWABLE ENERGY



UL Solutions has assessed
100+
UTILITY-SCALE SOLAR PROJECTS SINCE 2013

HOMER Energy Software is used in
190+
Countries, with over **250,000** users

1000+
PV products evaluations annually

100,000+
HYBRID POWER PROJECTS MODELED SINCE 2014

Forecast provider for
100+ GIGAWATTS
OF INSTALLED RENEWABLE ENERGY PROJECTS

Independent/Owners Engineer for
1000+
WIND AND SOLAR PROJECTS SINCE 2012

HOMER
Hybrid Optimization of Multiple Energy Resources

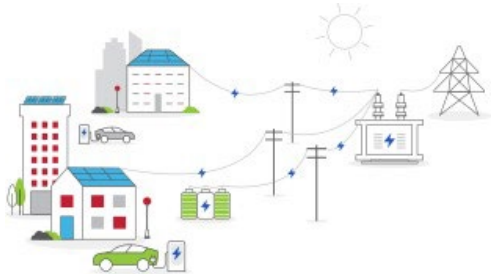
HOMER software models the technical and financial performance of your system

- Models key financial and performance parameters
- Minimizes costs or maximizes revenues
- Performs sensitivity and risk analysis for variable inputs
- Simulates real-world technical performance
- Is equipment and resource agnostic
- Delivers a choice of optimized designs

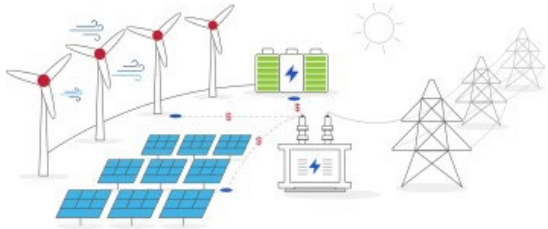
Across three categories of systems:



Standalone microgrids



Grid-tied, distributed generation



Utility-scale storage

Hybrid power system decisions simplified

FRONT-OF-THE-METER **BEHIND-THE-METER**

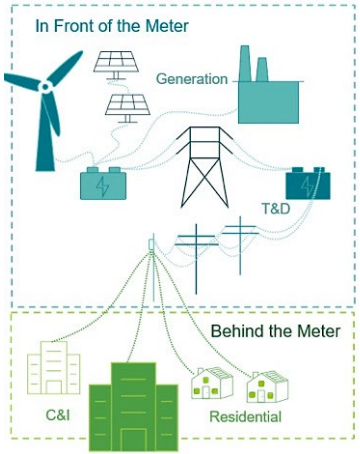
UTILITY-SCALE STORAGE | GRID-TIED DISTRIBUTED GENERATION | STANDALONE MICROGRIDS

HOMER Front

- Evaluate financial returns of energy storage, with or without solar and wind
- Maximize revenue and internal rate of return (IRR)
- Assess project feasibility
- Model storage dispatch, degradation and augmentation
- Validate revenue estimates of market participation and power purchase agreements (PPAs)

HOMER Grid

- Reduce energy costs for grid-tied systems that serve a local load



- Reduce demand charges
- Increase resilience
- Optimize electric vehicle charging
- Model combined heat and power
- Optimize against electric tariff structure
- Understand carbon and emission reductions

HOMER Pro

- Find the best energy mix and least-cost solution for powering off-grid systems for islands, mines, telecom towers, outposts
- Reduce energy costs, increase reliability and add renewables to your off-grid system
- Simulate and optimize any combination of energy resources
- Perform advanced sensitivity analysis

BATTERY SOLAR WIND GENERATOR FUEL CELL

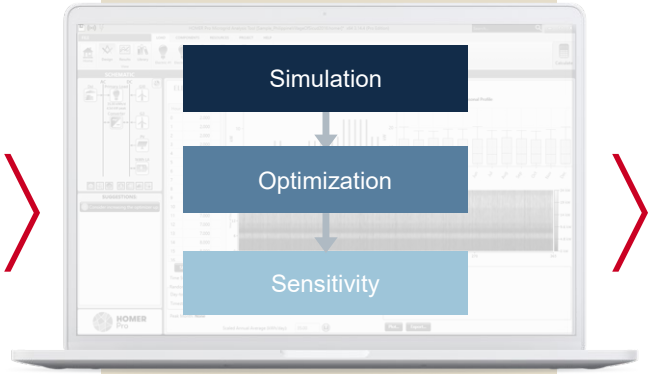
Performs techno-economic analysis for hybrid systems

INPUT

Project specific details

- Economics
- Load profile, revenue stream, market prices
- Site-specific renewable resources
- System components
- Capital expenditures (CAPEX) and operating expenses (OPEX)

ANALYSIS



Complex analytics, easily user controlled

RESULTS

Data-driven insights Informed decisions

- Financials
- System sizing
- Performance details
- Financials
- Emissions
- Various reports

Trusted global excellence in hybrid analysis

HOMER modeling sets the industry standard in over 193 countries

Developers

REC Solar, Shell, HITACHI, Hydro Tasmania, Shaw, CAT, SIEMENS, solarcentury, Mortenson, Kiewit, Sargent & Lundy

Component Suppliers

LEONICS, ABB, XANT, SAFT, Discover Battery, SPARKMETER, Panasonic, ennera, CAT, Sustainable Power Systems, Cummins, Rolls-Royce, Trojan Battery, EnSync Energy Systems, Schneider Electric, Spirae, Primus Power, REDT, CanadianSolar, TESLA

Education

IEA, IRENA, Massachusetts Institute of Technology, TUHH, UCSD, KTH, University of Cambridge

Engineering Firms

CADMUS, NAVIGANT, MRIGlobal, iED, HATCH, WSP, TETRA TECH, SOLENERGY SYSTEMS INC., ARUP, tecnalia, Intelligent Energy

Government & Financial Institutions

IFC, IDB, The World Bank, giz, ADB, UNDP, Rocky Mountain Institute

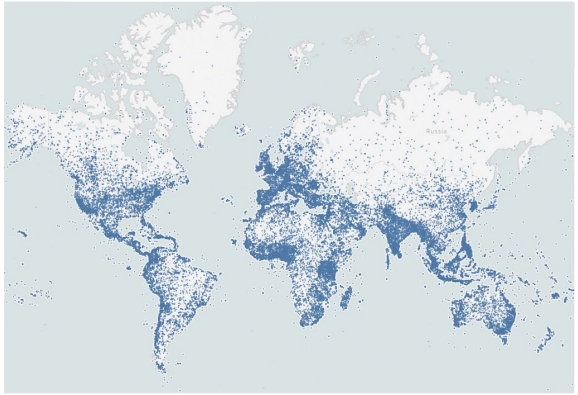
Utilities

Enel Green Power, ENGIE, e.on, New York Power Authority, nrg, Southern California Edison, conEdison, PEA, UTILACOR

Projects modeled worldwide

250,000 downloads, **90,000** projects modeled

3 million model runs, **193+** countries



HOMER Pro

Standalone microgrids



Standalone microgrids anywhere

HOMER Pro

Flexible microgrid design

Model any combination of microgrid equipment.

Economics

Simplify determining the lowest cost of energy when considering multiple technologies.

Climate

Include CO2 and other emissions reduction goals to understand the climate impact.

Evaluate risk

Sensitivity analysis for “what if” scenarios. What if fuel prices increase by 20% in five years?

Common applications

- Island nations and remote village
- Telecom sites
- Forward operating bases
- Mining operations
- Hydrogen

Equipment modeled

- Solar PV
- Generators (up to 20)
- Combined Heat and Power (CHP)
- Fuel cells
- Wind
- Hydro
- Biomass
- Hydrogen
- Storage



Robust microgrid modeling

Input

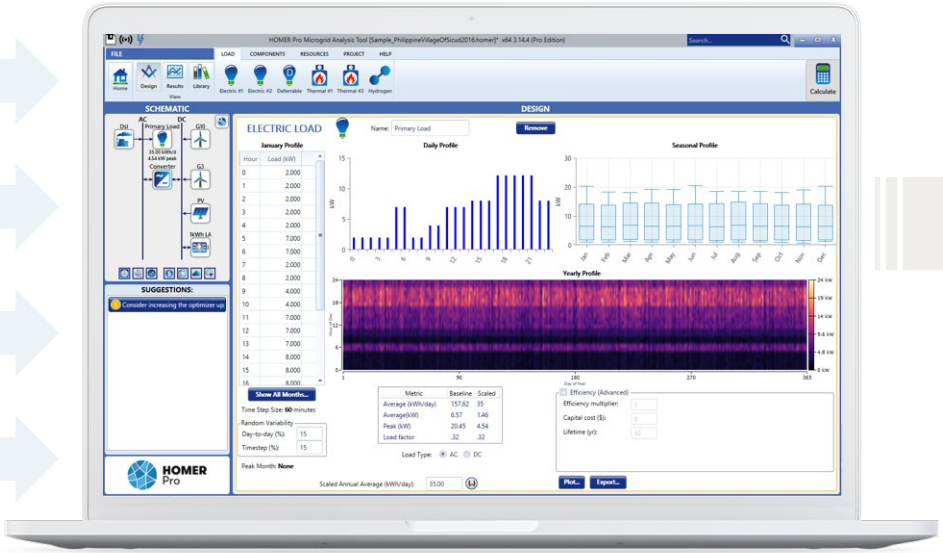
Load requirements
Electricity, thermal, hydrogen

Equipment options
Technology and size

System costs

Solar and wind resource data

HOMER Pro



Results

Least-cost optimized system

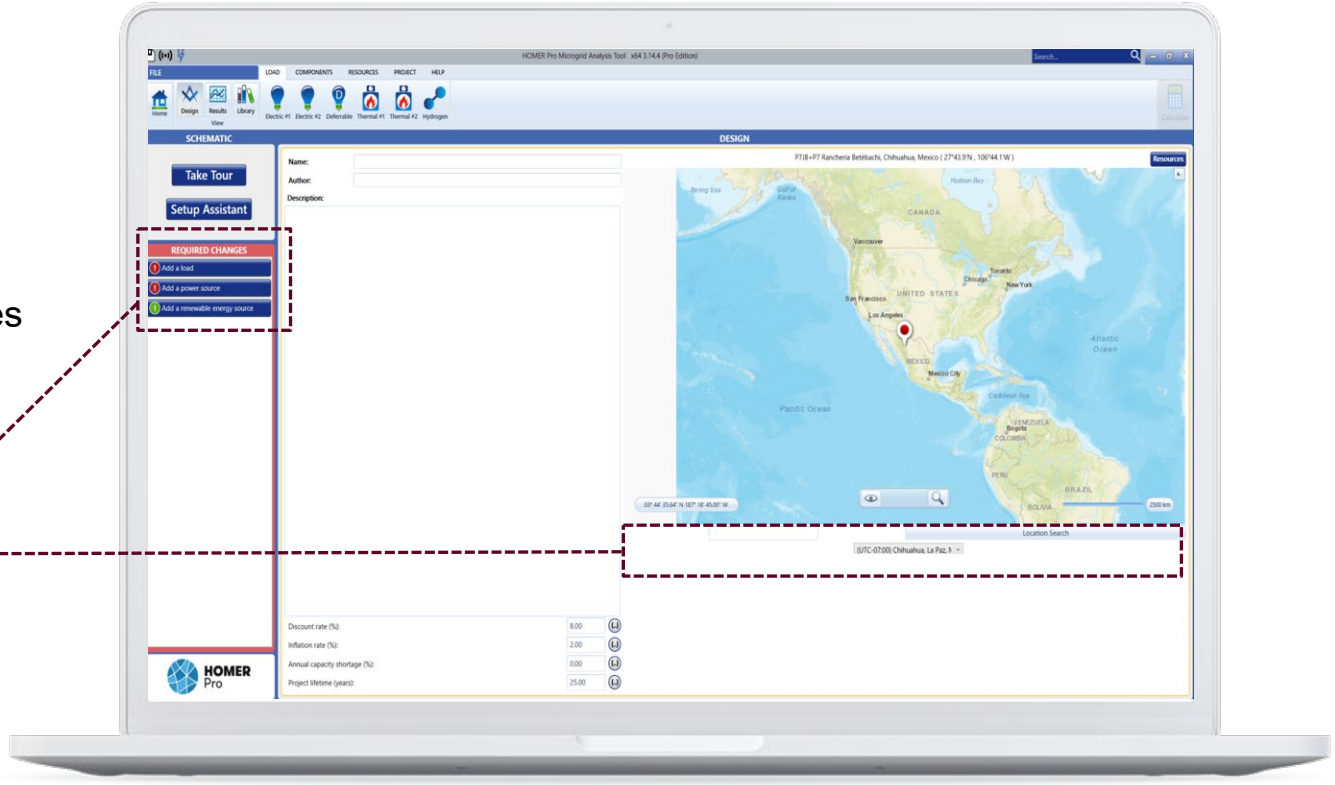
Sensitivity analysis

Software screen captures throughout are owned by UL Solutions

Model setup

HOMER Pro

- Define your location
- Download your resources
- Collect your data



Software screen captures throughout are owned by UL Solutions

1. Define the electric load

- **Define electric loads,** whether for an entire island, a facility or complex, or a building or system
- **Create synthetic loads** from available usage profiles or import your own load time series file
- **Set up multiple electric and thermal loads** for your system
- **Specify deferrable** and **non-deferrable loads**
- **Include thermal loads** for combined heat and power, as well as hydrogen loads for use as a reformer or an electrolyzer



Software screen captures throughout are owned by UL Solutions

2. Design system architecture and specify components

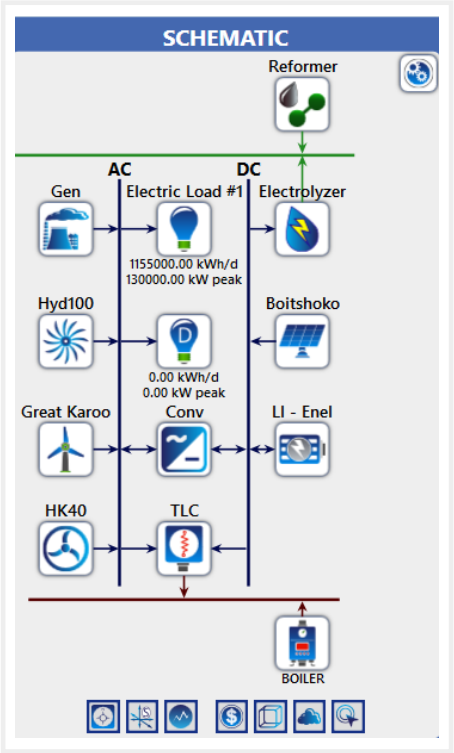
- **Define your system topology** and select the desired equipment for your specific application

  The system can be:

  - **Simple off-grid residence** or **more complex**, like an entire remote island system with multiple generators, technologies and loads

- **Choose the desired system architecture generators,** multiple generators, PV, wind, storage, thermal loads, hydro, hydrogen and even grid connections

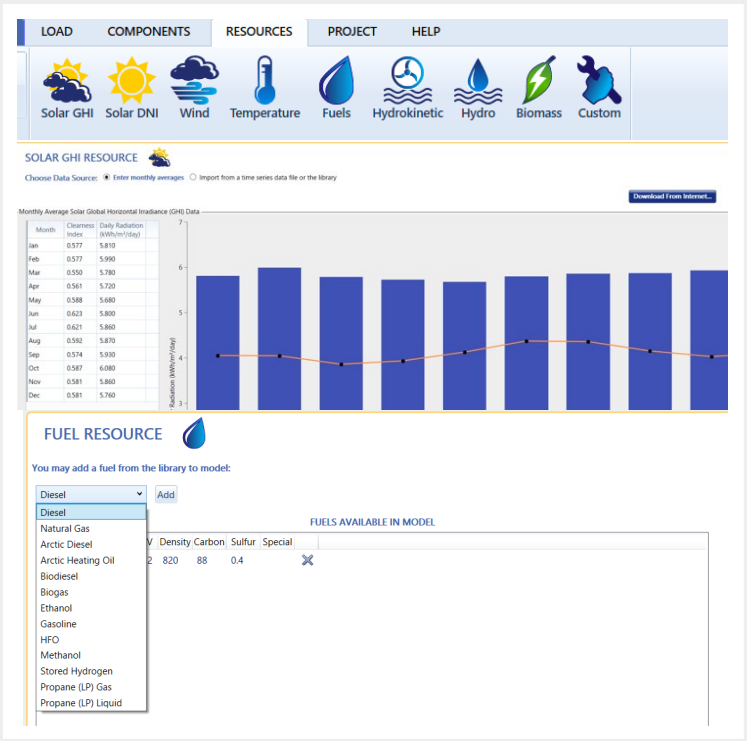
- **Set up your system controller:**
  - **Select from pre-defined dispatch strategies or define your own** strategy and control algorithms using C++ or MATLAB



Software screen captures throughout are owned by UL Solutions

3. Import resources (wind, solar, hydro) and define fuel costs

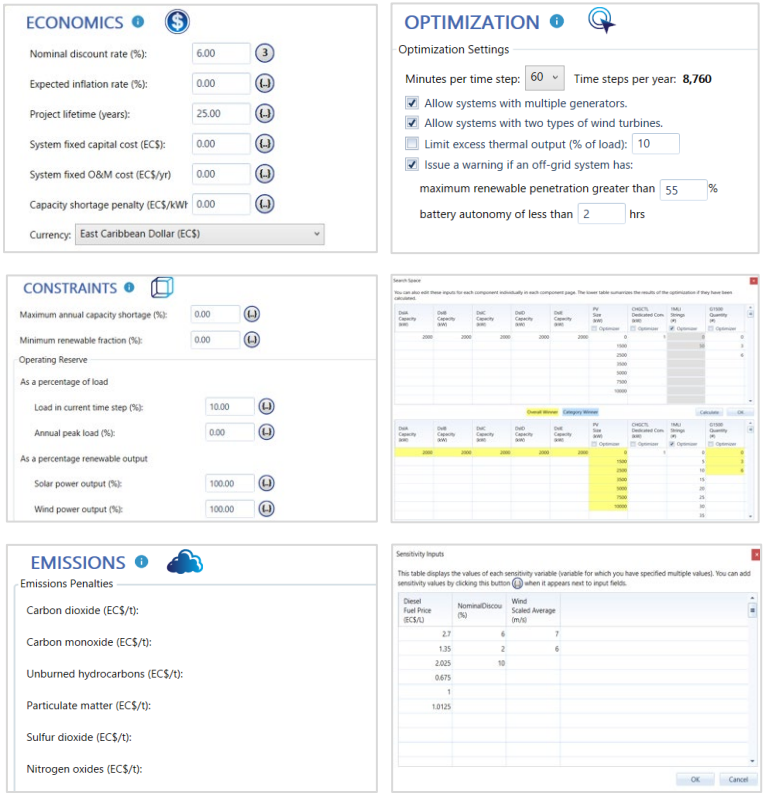
- **Import resource information** for your wind, solar, and hydro units:
  - **Import your own data or download publicly available** solar and wind resource information using the software for an initial estimate
- **Add temperature,** hydrokinetic and hydro information for run-of-river or hydro technologies
- **Add fuel data** including costs and emissions for biomass, oil, diesel, natural gas and other generator fuel sources



Software screen captures throughout are owned by UL Solutions

4. Finalize project information

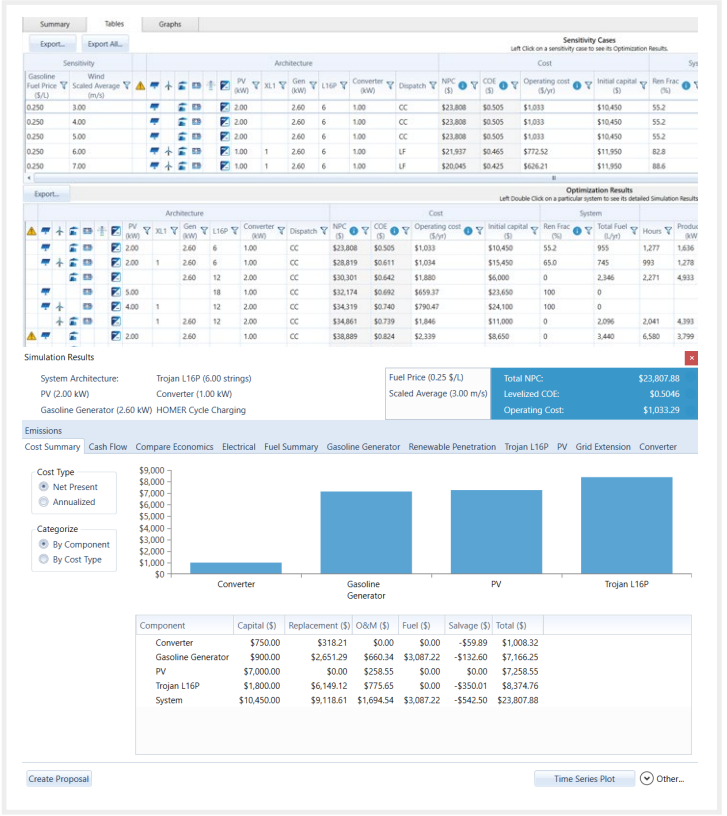
- **Define project economic constraints** for discount rate, inflation, project lifetime, fixed costs and operating and maintenance (O&M)
- **Define constraints for capacity shortage**, fraction of load served by renewables and peak loads your system must meet
- **Include penalties or price deterrents** for emissions, including carbon dioxide, carbon monoxide, hydrocarbons, particulate matter, sulfur dioxide and nitrogen
- **Set up optimization parameters** and set boundaries on your search space
- **Establish sensitivity parameters**
- **Define multi-year analysis** inputs such as escalators
- **Save** and **run the analysis**



Software screen captures throughout are owned by UL Solutions

5. Analyze results

- **View results** across thousands of simulations
- **Analyze the winning or least-cost system** or prioritize by another metric
- **Evaluate the project cost summary,** cash flow, economics and renewable penetration levels
- **Review costs** and cash flows by component
- **Drill down into the dispatch details** by using the time series viewer
- **Assess** the fuel, electrical and heat usage
- **Compare emissions** between different system configurations and integration of renewable energy



Software screen captures throughout are owned by UL Solutions

HOMER Pro

Microgrids

Case studies



International gold producer turns to clean energy

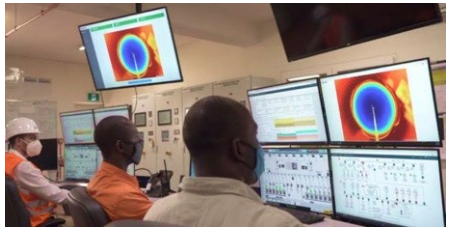
HOMER Pro models the large off-grid mining hybrid power system

Client	**BayWa r.e., global renewable energy company Munich, Germany**
Problem Integrate large renewable energy plant into a constrained grid	• B2Gold's off-grid Fekola Gold Mine in Mali, West Africa, powered operations with heavy oil generators in an onsite thermal power plant. • B2Gold wanted to improve efficiencies, lower operating costs and support decarbonization, local job creation and sustainability. • Integrating a large renewable energy plant into a small, isolated, constrained grid is challenging.
Solution Design the right-size solar-plus-storage system	• Renewable energy companies Suntrace and BayWa r.e. designed a solar plus storage system integrated into the existing power plant. • System successfully commissioned in March 2021 and has operated at full solar plant capacity since July 2021. • Energy costs lowered and B2Gold partially protected from oil price increases.
HOMER Engagement HOMER Pro simulated various systems and Matlab link enabled coding of custom controller	• HOMER Pro simulated systems with and without storage, then BayWa r.e. outlined solar plant options, selected the best approach and performed detailed modeling. • HOMER Pro's Matlab link enabled coding of a custom energy management system, modeled component sizes and optimized the system configuration. • The feasibility study's control concept was implemented in the energy management system deployed onsite by battery vendor Wärtsilä.

HOMER Pro

BayWa r.e.





KiloWatts for Humanity empowers Zambia

HOMER Pro sizes microgrids for solar kiosks

Client	**KiloWatts for Humanity, non-profit organization Seattle, Washington**
Problem Provide energy access to remote, rural Africa	• In the south-central African country of Zambia, the electrification rate of rural areas is only 11%. • KiloWatts for Humanities vision is to bring first-time electricity access to places that most need it, with a focus on renewable energy solutions to "change lives in a community."
Solution Build solar kiosks	Solar kiosks are small buildings equipped with renewable energy to serve • Specific business purposes, such as a shop or service like a barbershop. • Community need for battery and phone charging stations.
HOMER Engagement Size microgrids to serve electric load of each solar kiosk	• KiloWatts for Humanities relies on HOMER Pro to size the microgrid to meet the community's exact electrical needs. • Each project has similar load profiles, but its own specifics, such as the addition of a water pump or welding tools. • UL Solutions donates the HOMER Pro license to the organization.

HOMER Pro

KiloWatts for Humanity





Belize seeks clean energy independence

HOMER Pro software models a variety of technologies and system sizes

Client

Rocky Mountain Institute (RMI), an independent, non-partisan non-profit. RMI works with businesses, policymakers, communities and other organizations to identify and scale energy system interventions that will cut greenhouse gas emissions at least 50% by 2030.

Problem

Create a strategy for energy independence with 95% renewable energy for Belize by 2030

- Belize buys about 70% of its energy from IPPs and the Mexican government.
- In 2012 the government of Belize endorsed the National Energy Plan to achieve clean energy independence and be 95% renewable by 2030, which required a sustainable energy strategy to achieve its ambitious goals.

Solution

Work with a team of experts to plan and develop energy systems and train locals

- RMI played a key role in developing a plan towards Belize's renewable energy expansion with the help of a HOMER expert.
- The team assessed existing biomass and hydro plants, planned future expansions, set the needed timeframes to achieve low-carbon communities by 2033, helped with the design of solar and wind systems, created energy efficiency programs and trained Belize's professionals on the design process.

HOMER Engagement

Provide professional services for design and planning using HOMER Pro. Train local professionals on the design process

- RMI and UL Solutions worked together to create a year-by-year plan on the implementation of different clean energy technologies that the Belize government could follow to achieve its clean energy goals.
- The analysis was performed using HOMER Pro.
- UL Solutions also trained the local utility company, policymakers and others on the design process of a variety of system architectures. This included biomass, hydro, power purchase agreements, solar, storage and wind systems, among others.

HOMER Pro

RMI



UL Solutions

HOMER Grid

Distributed generation



Grid-connected distributed generation

HOMER Grid

More than solar and storage

Model any technology found in a distributed energy resources (DER) environment.

Value stacking

Evaluate solutions that increase value: demand charge reduction, demand response, capacity or production incentives, self-consumption and CO2 savings in carbon reduction programs.

Easy-to-use utility tariff and component database*

Save hours of setup with a built-in tariff database or build your own tariffs and choose from hundreds of OEM components. Specifically for complicated tariffs and incentives.

Resiliency, blackouts and decarbonization

Model critical and non-critical loads separately and provide backup options for a part of or the entire facility.

EV charging

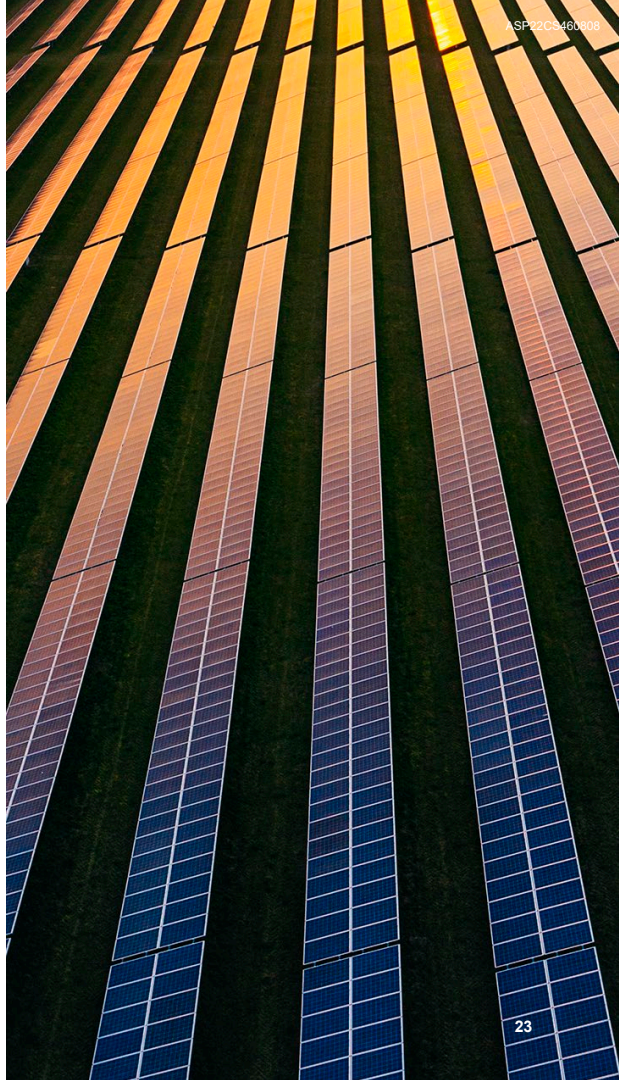
Avoid demand charges for single or multiple charging stations.

Common applications

- Hospitals
- Data centers
- Commercial buildings
- University campuses
- Hotels
- EV charging stations

Equipment modeled

- Solar PV
- Generators (up to 20)
- CHP
- Fuel cells
- Wind
- Storage



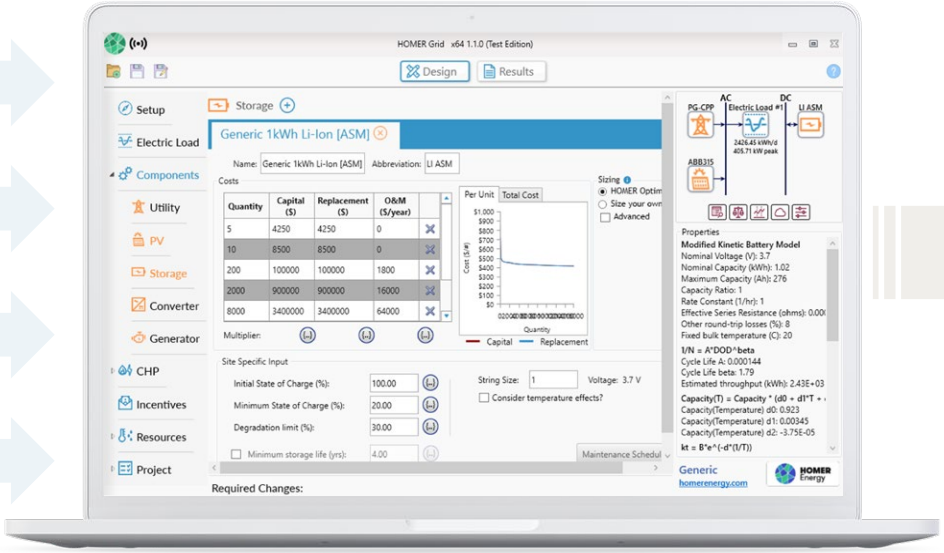
* Tariff database currently available in U.S., Canada and part of Mexico

Help maximize value through DER optimization

Input

- **Tariffs**
- **System components**
- **Load profile**
- **Solar and wind resource data**

HOMER Grid



Results

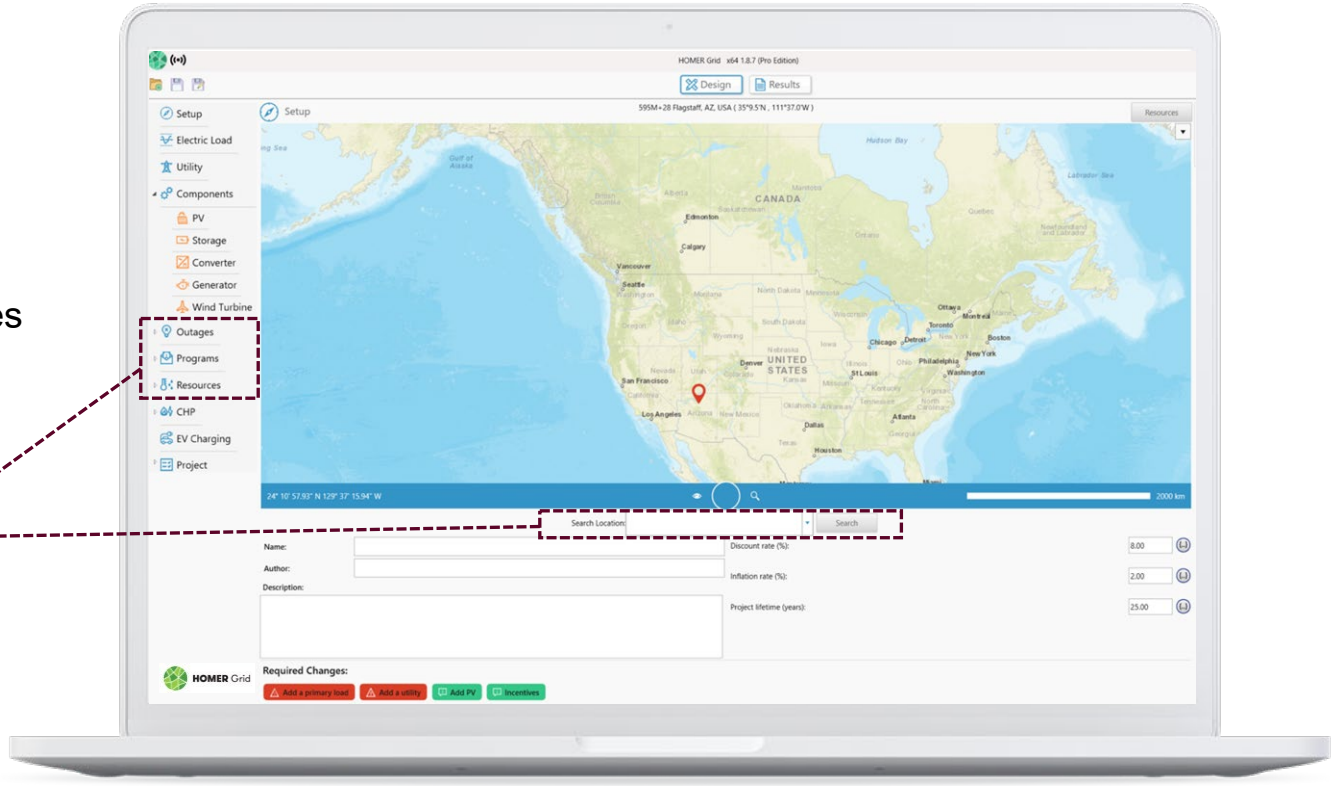
- **Optimized IRR, ROI and savings**
- **Reduced costs and carbon**

Software screen captures throughout are owned by UL Solutions

Model setup

HOMER Grid

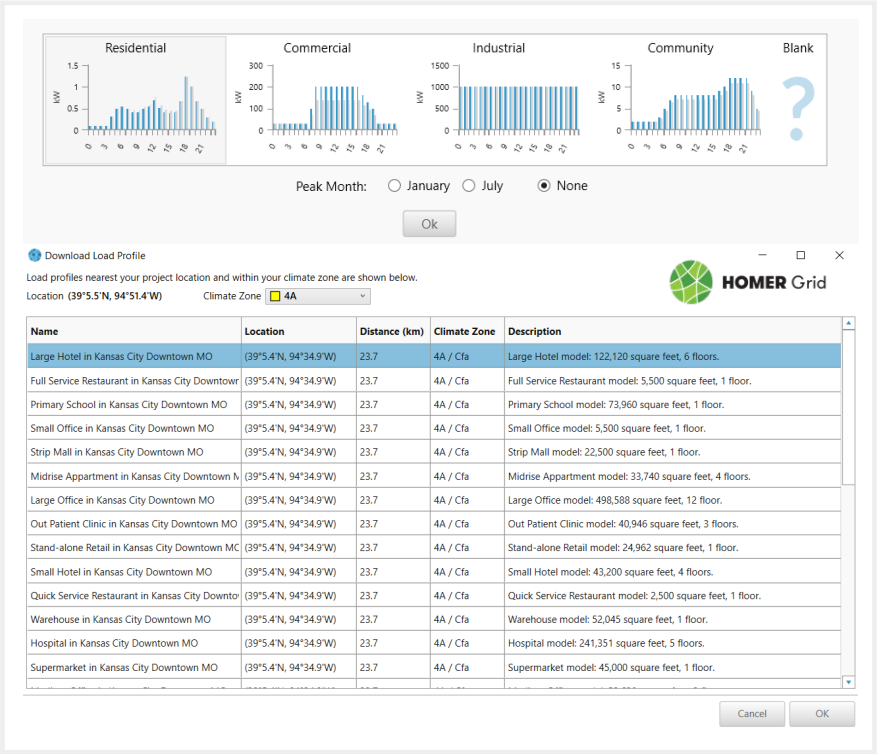
- Define your location
- Download your resources
- Collect your data



Software screen captures throughout are owned by UL Solutions

1. Define the electric load

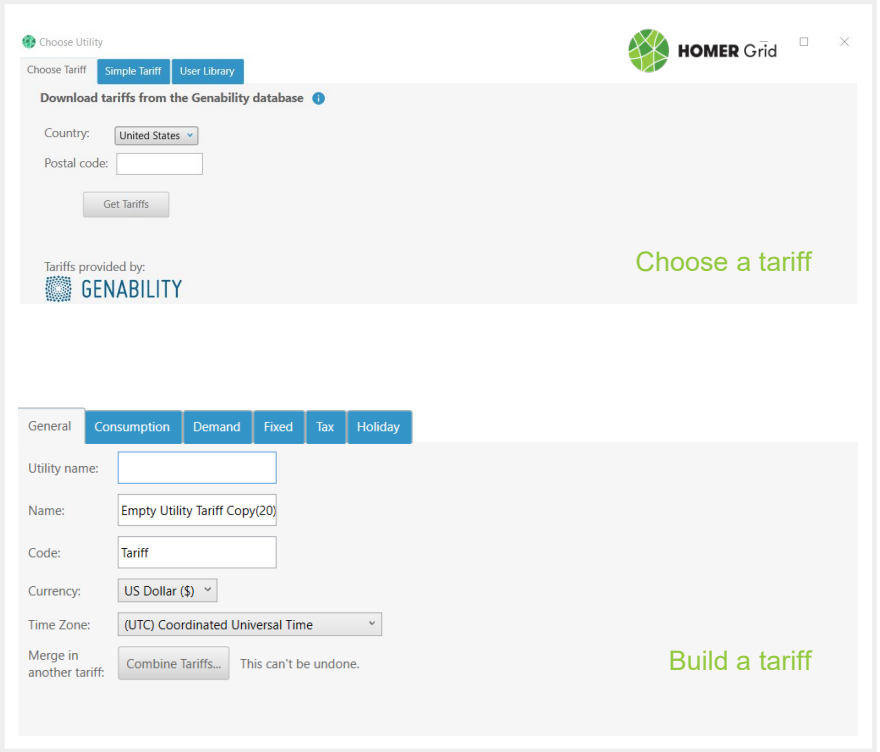
- **Define** your electric load for your building, campus or system.
- **Use** the open database for thousands of load profiles for various facilities in different climate zones.
- **Import** your own load data via a time series file from electric bills.
- **Create** a synthetic load from a profile for different building types.
- **Define** loads as critical or non-critical for servicing during outages or demand response events.



Software screen captures throughout are owned by UL Solutions

2. Set the utility tariff and rate structure

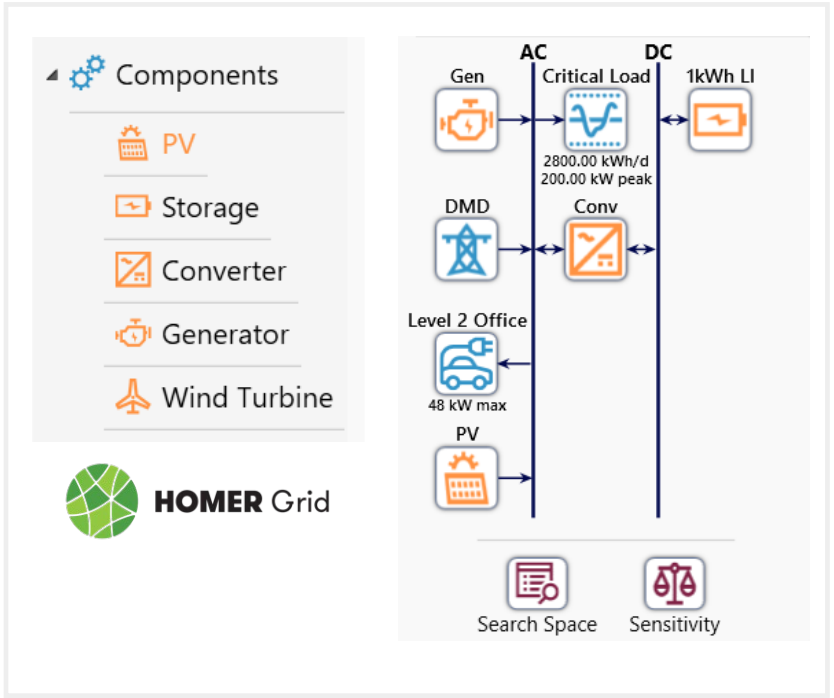
- **Model** tariff structures available from your utility: energy costs and demand charge rate structures.
- **Choose** tariffs from our tariff library or create your own tariffs using our simple or advanced tariff builder.
- **Combine** two separate tariffs for a single facility with multiple energy meters.
- **Compare** different tariffs in a sensitivity analysis to find the optimal tariff for your project.



Software screen captures throughout are owned by UL Solutions

3. Select your desired distributed generation resources

- **Consider** solar, storage, different converter architectures for AC and DC coupled systems, generators, electric vehicle (EV) charging and combined heat and power (CHP).
- **Add costs** for capital and replacement and O&M rates for equipment.
- **Model** energy costs and demand charge rate structures.
- **Import PV** data from PVsyst, HelioScope, your own production files or choose a PV module and let HOMER model the energy production for you.
- **Choose your energy storage technology** and assess using our ideal or advanced energy storage models.
- **Set the upper and lower bounds** for desired system sizes, specify different discrete sizes, or let HOMER optimize the system configuration and size.



Software screen captures throughout are owned by UL Solutions

4. Consider combined heat and power

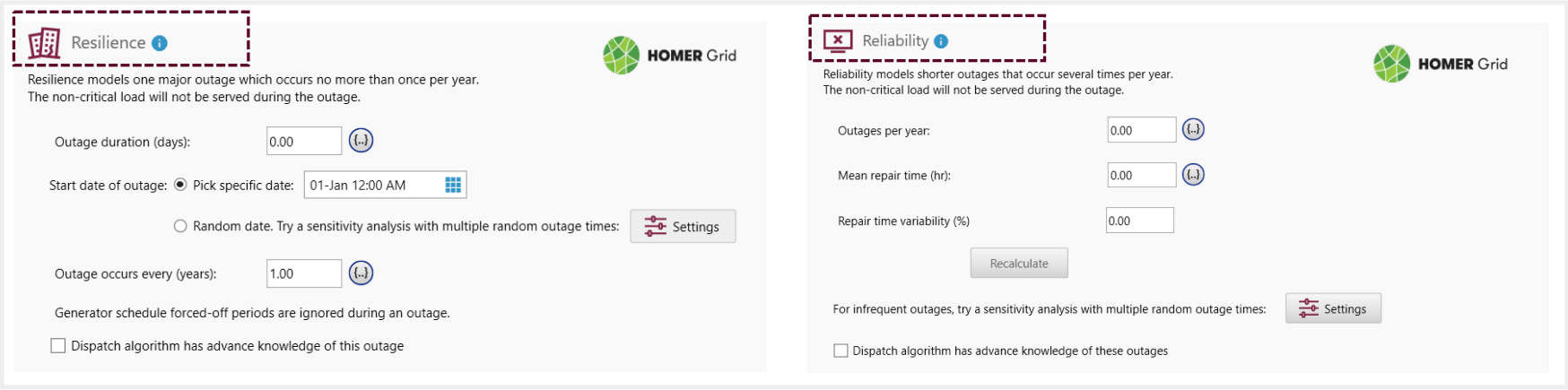
- **Specify** your thermal load:
  - Import a time series load
  - Access our load database
  - Create a synthetic load from a profile
- **Select** and **add** a boiler and heater and define:
  - Fuel source
  - Costs
  - Emissions



Software screen captures throughout are owned by UL Solutions

5. Specify resilience and reliability requirements

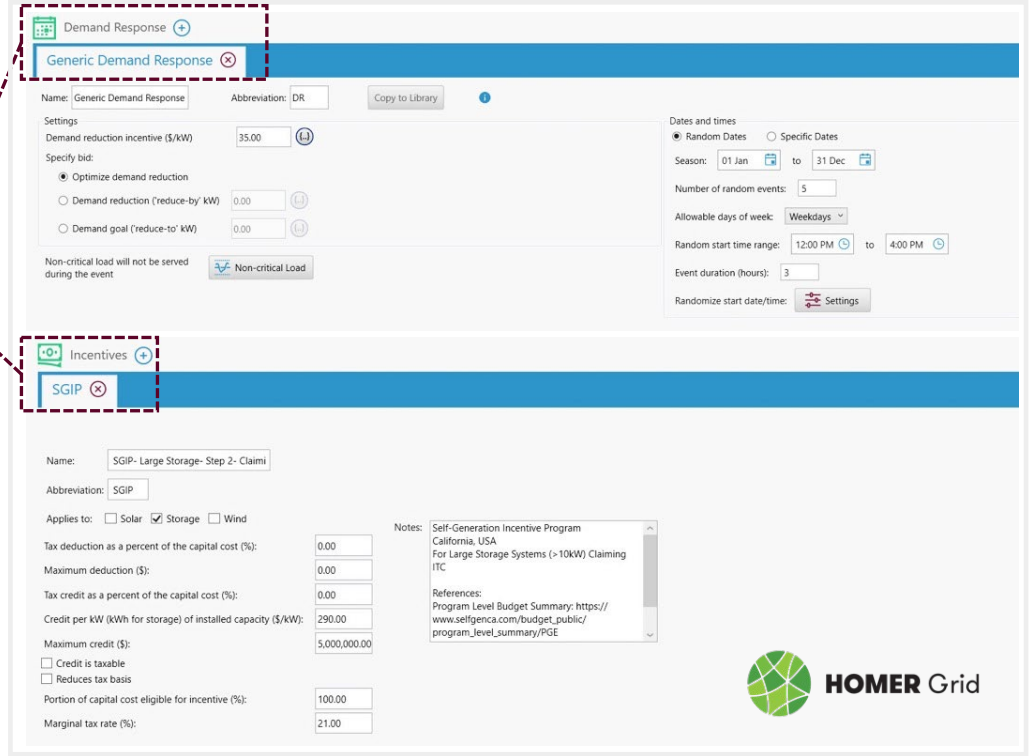
- **Model** required resilience or reliability for peak energy periods or grid outages.
- **Specify** critical and non-critical loads for your facility.
- **Model outage** events to determine the system's resilience and reliability.
- **Specify outage settings** and system constraints or required operating reserves during outages.
- **Prioritize** the importance of resilience and reliability versus cost savings such as peak shaving.
- **Understand** the trade-off between enhanced resilience/reliability and system costs.



Software screen captures throughout are owned by UL Solutions

6. Specify demand response programs or incentives

- **Include eligible demand** response programs
- **Include system incentives**
- **Specify other incentives,** including:
  - Bonus depreciation
  - Investment tax credit
  - Modified accelerated cost recovery system (MACRS)
  - Capacity- or production-based incentives



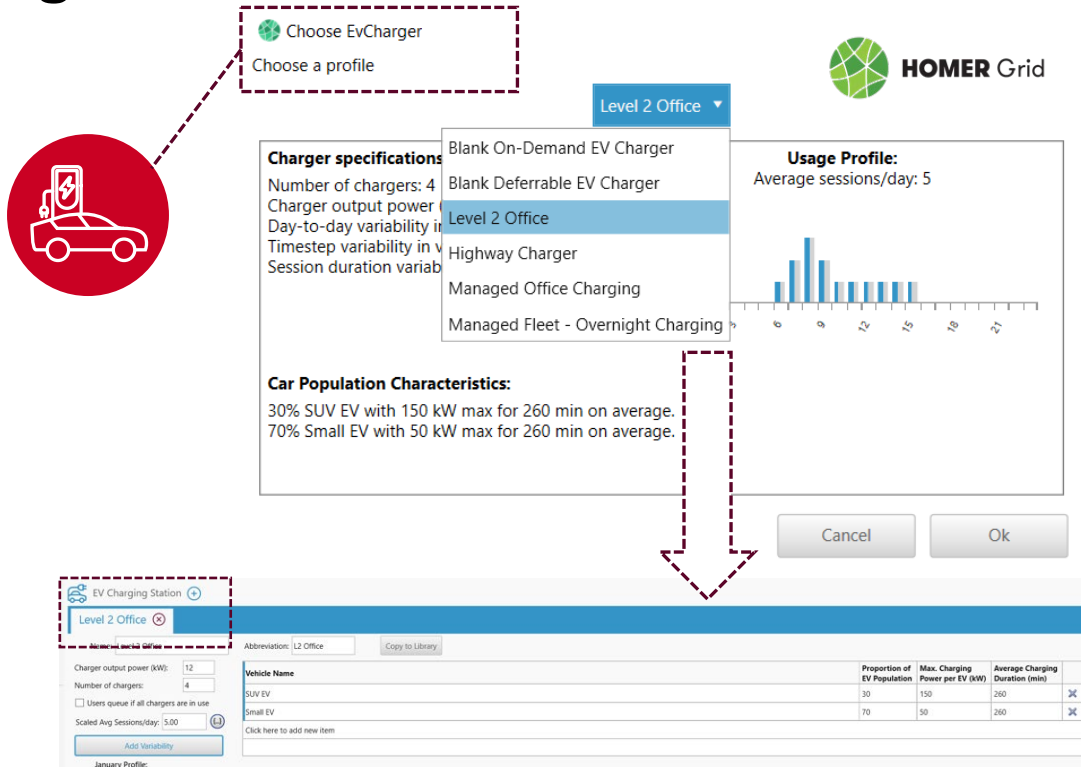
Software screen captures throughout are owned by UL Solutions

6. Consider EV charging

EV charging impacts electric loads, energy costs and demand charges of your system.

- **Select your charging types:**
  - Fleet charging
  - Office or residential use
  - Highway charging

- **Model EV variables**:
  - Type and number of electric vehicles
  - Introduce variability of month, season or time of day



Software screen captures throughout are owned by UL Solutions

6. Understand your results

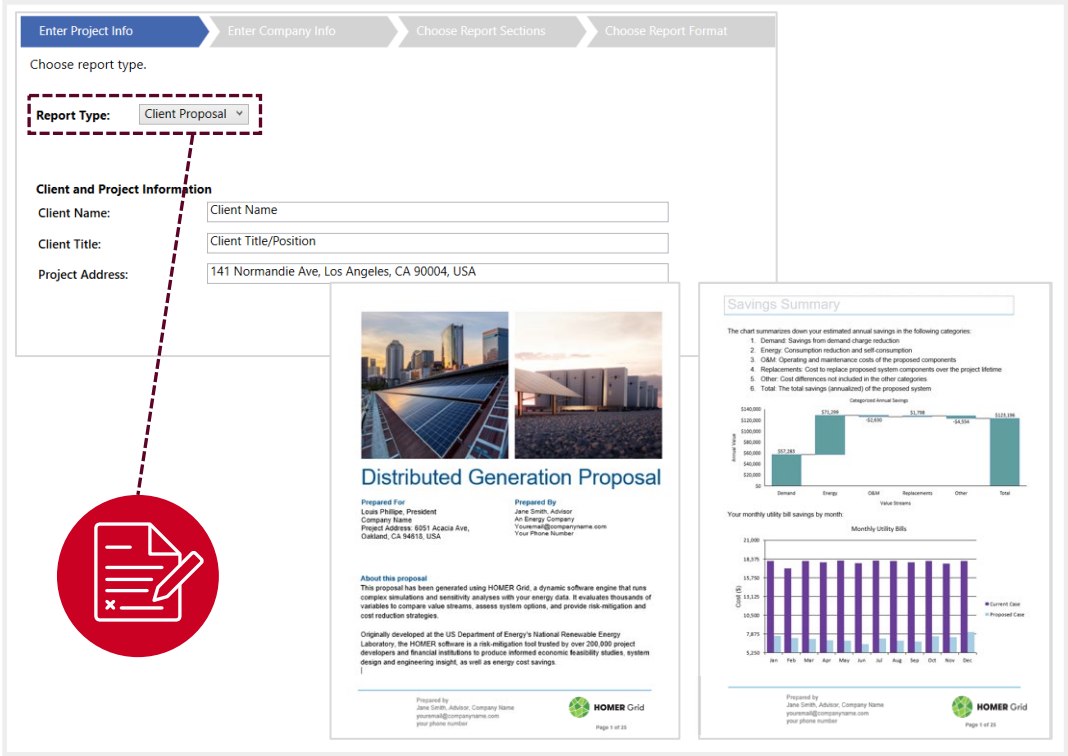
- **Determine optimal system sizing** to meet your needs.
- **Understand** project economics and cash flow.
- **Perform sensitivity analysis** to understand the impacts on the project financials.
- **Review performance** over the project lifetime by year.
- **Drill down** to time-step by time-step granularity to understand the behavior of the system.



Software screen captures throughout are owned by UL Solutions

7. Easily generate a report with our proposal writer

- **Output** a customized proposal report with one click.
- **Communicate** with stakeholders, clients and project team members.
- **Present key** aspects of your analysis, including cost comparisons and economic value streams.
- **Add** your company logo and customize with your brand.



Software screen captures throughout are owned by UL Solutions

HOMER Grid

Value stacking distributed energy resources (DER)

Case studies



Yarmouth microgrid delivers resilience and cost savings

HOMER® Grid models optimal hybrid power system and demand response incentives

HOMER Grid

RIVERMOOR ENERGY

Client

Rivermoor Energy, Boston, Massachusetts
Rivermoor Energy worked with engineering partner Weston & Samson

Problem

Provide energy resiliency to critical municipal operations during power outages for coastal town

- Faced with frequent extreme weather, the Cape Cod town of Yarmouth needed to power critical facilities during prolonged power outages.
- Town officials received a Massachusetts Municipal Vulnerability Preparedness (MVP) grant and selected Rivermoor Energy to develop a clean energy resiliency plan for two mission-critical facilities. Requirements included cutting energy costs and carbon emissions and serving as a blueprint for other Massachusetts towns.

Solution

Microgrid to power new Emergency Operations Center with clean, reliable energy and cost savings

- The new microgrid includes a 264-kilowatt (kW) PV array and a 180-kWh battery storage system with backup generators.
- The microgrid provides +90% reduction in greenhouse gas emissions of firehouse operations and cuts energy costs and peak demand via optimized dispatch strategies.
- The microgrid participates in a demand response program to reduce electricity use with energy storage during summer and winter peak-demand periods.

HOMER Engagement

Provide engineering support to model a grid-tied system to meet the project's goals using UL Solutions HOMER Grid

- Rivermoor Energy retained UL Solutions to model a clean, reliable system to power the fire station and its Emergency Operations Center and a regional septic waste center.
- The team used HOMER Grid to analyze the grid-tied system, capable of operating in islanded mode during emergencies.
- UL Solutions' modeling and analytics helped Rivermoor Energy formulate the business model for the projects to benefit the town of Yarmouth and for the planned project financing.



Project	Location	Technologies	Goals achieved
PV Solar + Energy storage	Yarmouth, Mass., USA	PV 264 kW Storage: 180 kWh Microgid controller (Existing backup generator)	Resiliency GHG reduction Energy cost reduction Demand response

UL Solutions

Financial impact of EV charging stations

California hospital combines profitability, purpose and power.

Client

Hospital/ Medical Center, Oakland, California

Problem

High operational cost of decarbonizing ambulance fleet

- Ageing ambulance fleet needs to be upgraded to electric vehicles to reduce carbon emissions and provide services without compromising patient health.
- Electrification of ambulance fleet resulting in high energy cost.

Solution

Integrating solar PV plus energy storage to maximize savings and revenues

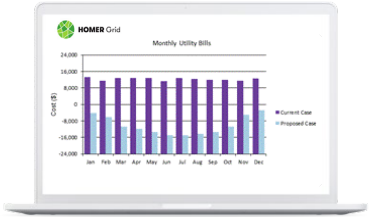
- Integrating solar carports with capacity of 1,000 kW PV and Storage of 500kW/500 kWh system yields the following results:
  - Initial equity investment – $2.4 million
  - 20-year internal rate of return – 15.7%
  - Payback – 6.2 years

HOMER Engagement

HOMER® Grid enabled design of innovative microgrid.

- UL Solutions HOMER® Grid software was used to determine the best mix of resources for the least cost solution by calculating the value of demand charge reduction, energy arbitrage and self-consumption.
- HOMER Grid analysis determined the most efficient component capacities to minimize cost and maximize revenues through the specific electric tariff for EV charging station coupled with solar PV plus energy storage.

HOMER Grid





Financial results	
10.8 yr	Simple payback
5.25%	Return on investment
7.87%	Internal rate of return
$699,568	Net present value
$3.57M	Capital investment
$330,015	Annualized savings

What is the cost of an outage?

For this Colorado car dealership, the cost of a power outage was unrecoverable.

Client	**Ehrlich Toyota car dealership, Greeley, Colorado**
Problem Recurring power outages cause significant revenue loss	• Between two and four prolonged power outages annually caused interruptions and significant financial losses. • When the power goes down, computers can't process transactions and business comes to a standstill. Customers who leave before completing their car purchase may not return.
Solution Island from the grid during outages with a renewable solar-plus-storage system	• IPOWER Alliance designed a renewable grid-tied hybrid power system. • Dealership can island power from the utility and run all day on solar plus storage and for several hours without sun. • Dealership operation continued during two power outages on Black Friday.
HOMER Engagement HOMER Grid enabled the design of an innovative microgrid.	• HOMER Grid software was used by Ageto Energy to model the 100% renewable grid-tied system. • HOMER Grid analysis enabled Ageto to determine the most efficient innovative microgrid that integrated a completed LED lighting upgrade, a 368 kW solar array on a custom carport plus 228 kWh battery storage and an Ageto Energy control system.

HOMER Grid

Ehrlich TOYOTA SINCE 1946





HOMER Front

Utility-scale storage



Front of the meter – storage, wind and solar

HOMER Front

Utility-scale generation

Model any combination of microgrid equipment.

Battery management

Dispatch based on hourly and sub-hourly price forecasts to model energy-shifting arbitrage and capacity market participation.

Multiple revenue streams

Validate revenue projections of power purchase agreements, whole energy participation (day-ahead and real-time) and capacity market participation.

System sizing

System sizing and optimization for power purchase agreements (PPA), request for proposal (RFP) and information assurance (IA) compliance.

Augmentation strategies

Advanced storage module to model degradation and augmentation strategies in use cases, including cycling, depth of discharge, throughput and time horizon.



Robust hybrid utility-scale front-of-the-meter modeling

Input

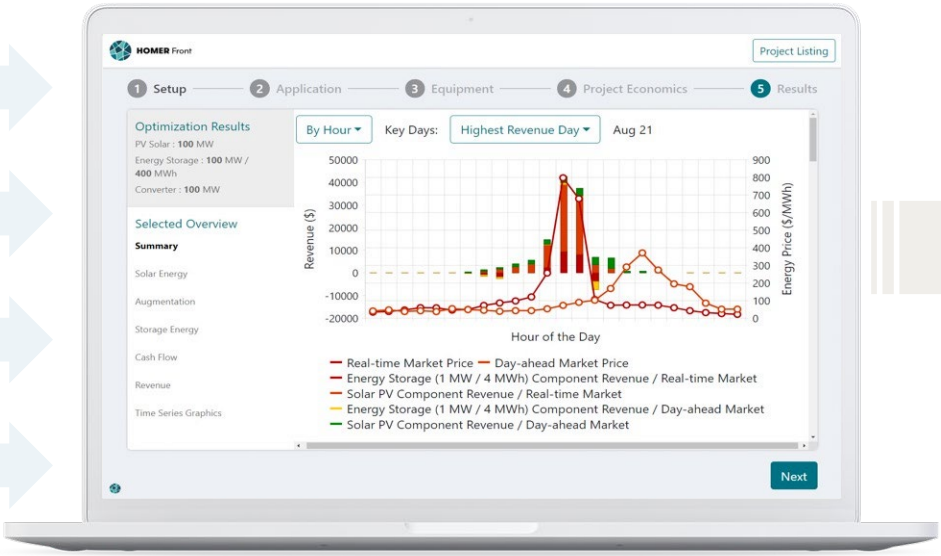
Energy and capacity markets prices, power purchase agreements

Equipment size

Incentives

Solar and wind production data

HOMER Front



Results

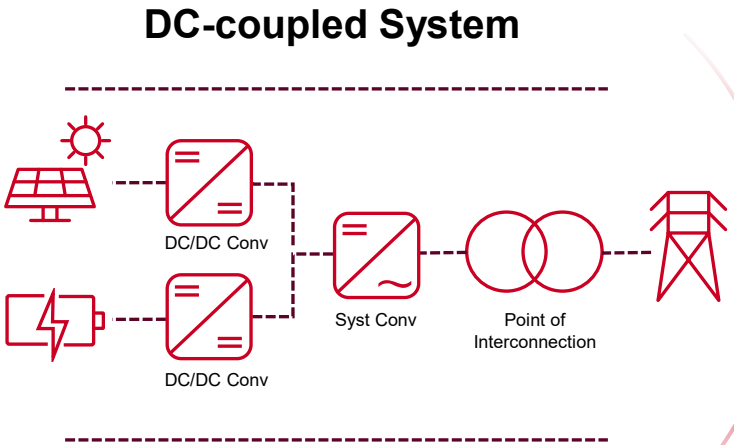
Maximum revenue and IRR

Sensitivity analysis

Optimization results

Software screen captures throughout are owned by UL Solutions

Key capabilities: accurately model markets and operation



Integrate energy market revenue
hourly and sub-hourly locational marginal pricing

Model capacity market revenue
storage system compensated while held in reserve and when discharged

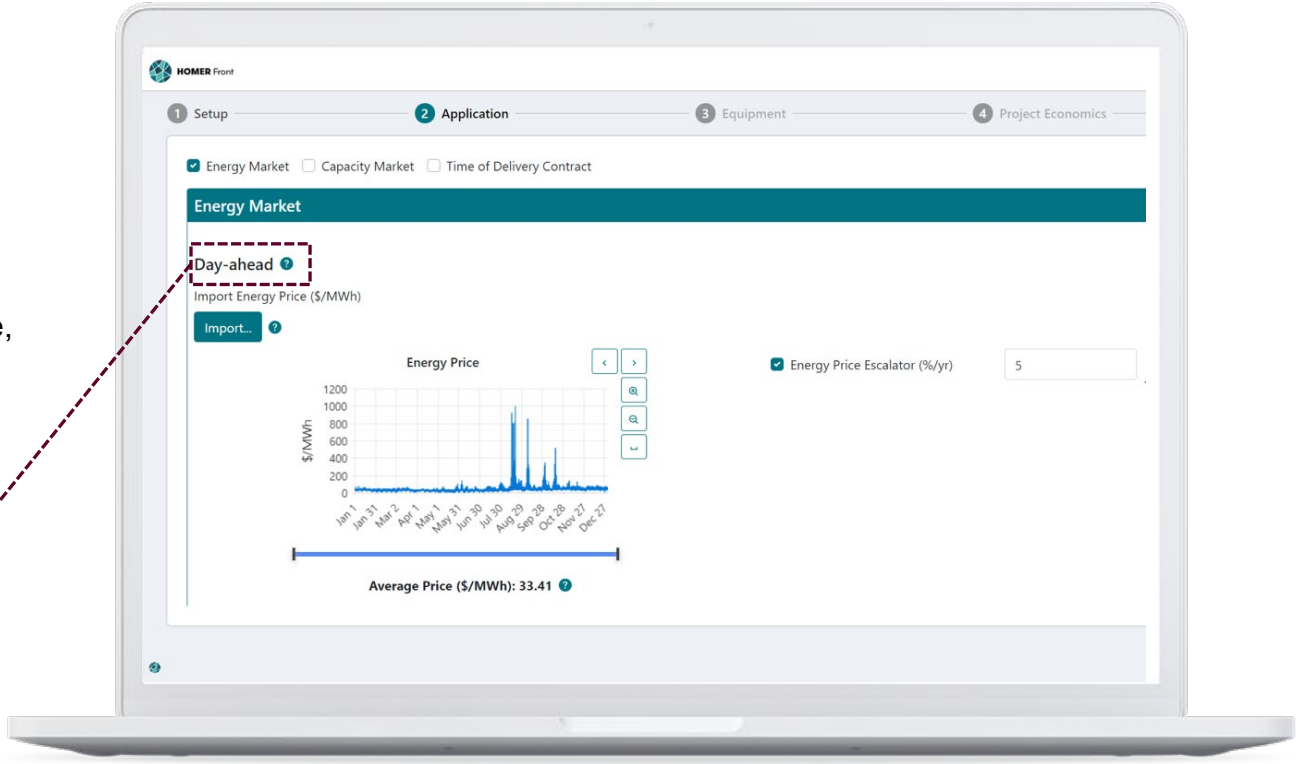
Develop strategies
for capacity retention and augmentation

Import and adjust various formats
for solar and wind production and resource data

Model setup – energy markets

HOMER Front

- **Energy market inputs**
  - Wholesale energy: day-ahead and real time, hourly and sub-hourly
  - Locational marginal pricing (price strips)

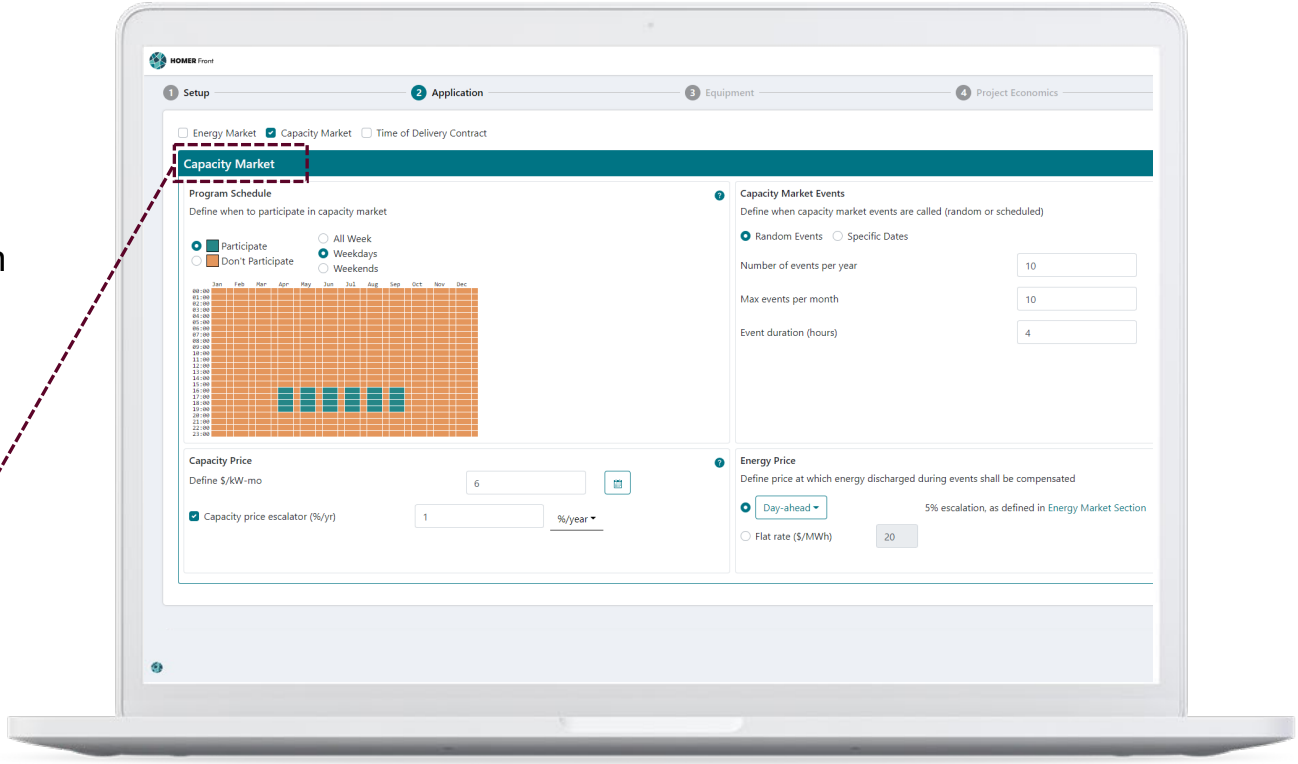


Software screen captures throughout are owned by UL Solutions

Model setup – capacity market and other revenue streams

HOMER Front

- **Supports** renumeration for reserved capacity and additional revenue opportunity for dispatched energy

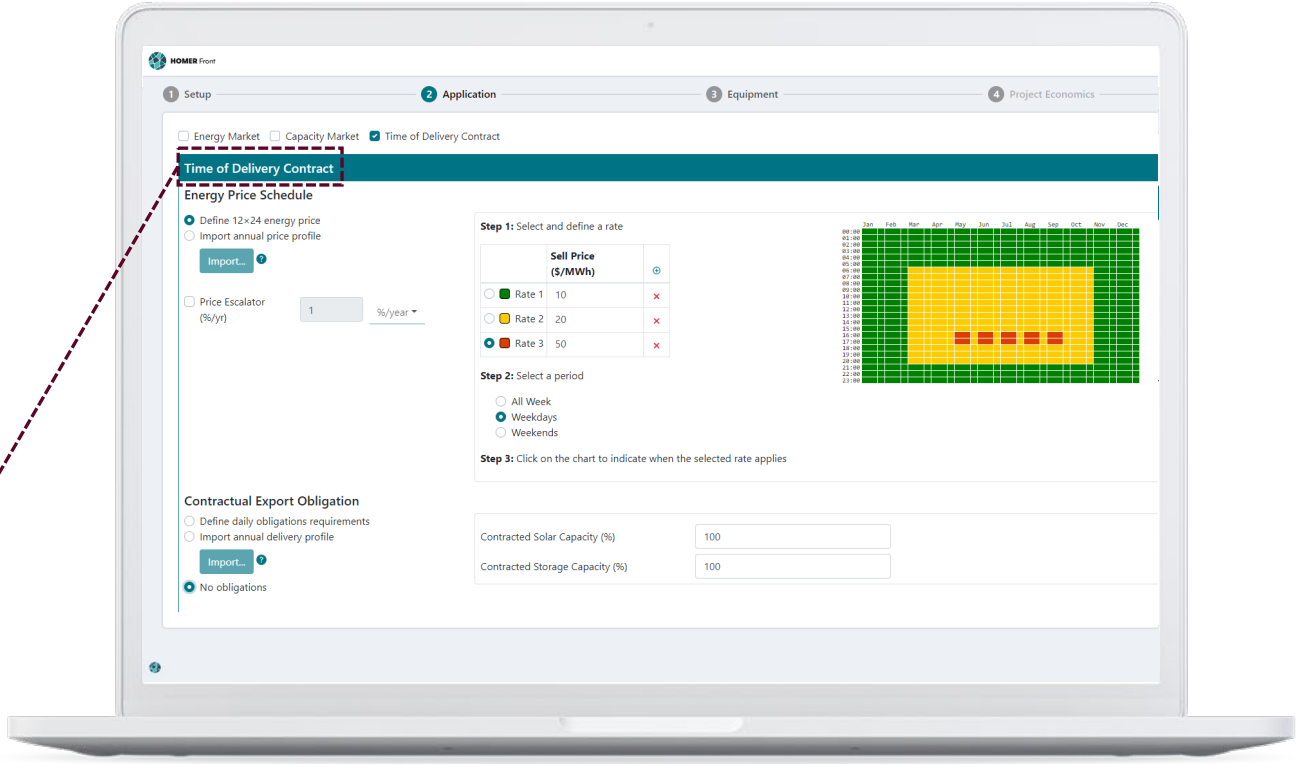


Software screen captures throughout are owned by UL Solutions

Model setup – power purchase agreements, time of delivery

HOMER Front

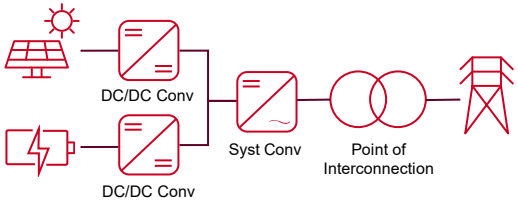
- **Contracted** solar and storage capacity may be compensated at a structured rate



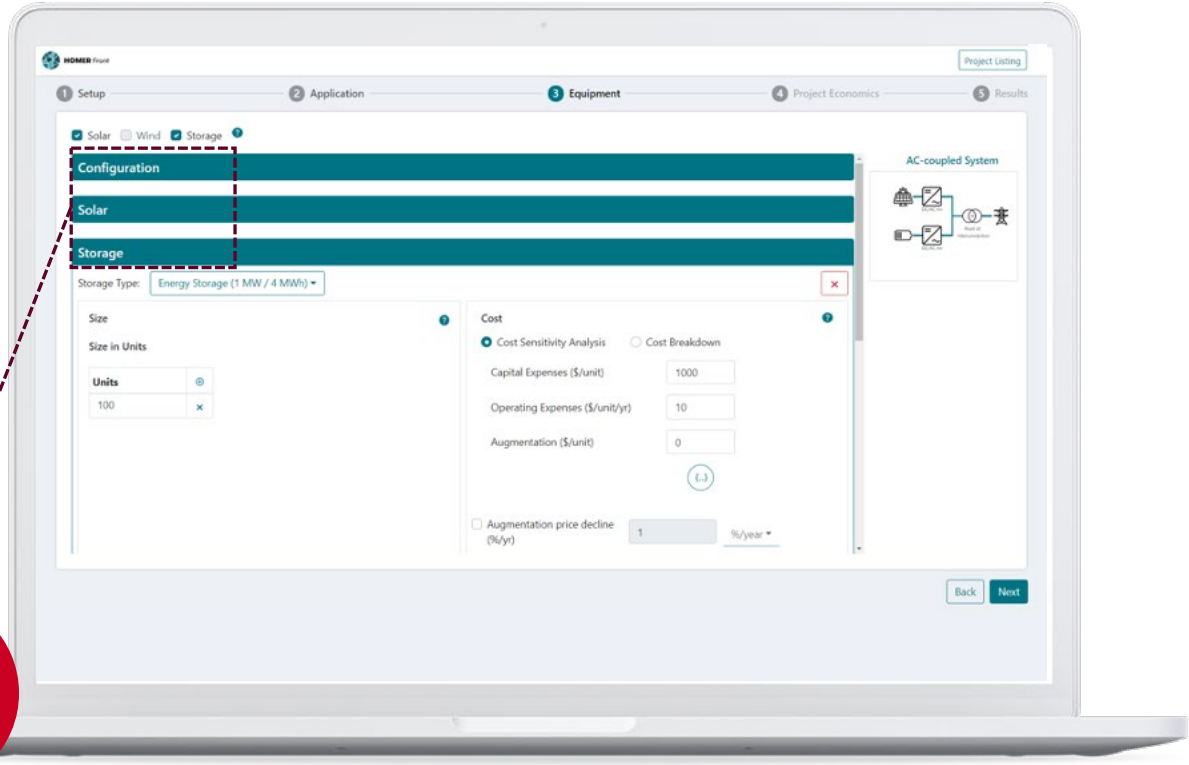
Software screen captures throughout are owned by UL Solutions

Model setup – system size and cost, augmentation strategies

DC-coupled System



- **Optimally size and cost system architecture:**
  - Configure solar and storage equipment
  - Input equipment pricing
  - Test augmentation strategies with robust storage modeling

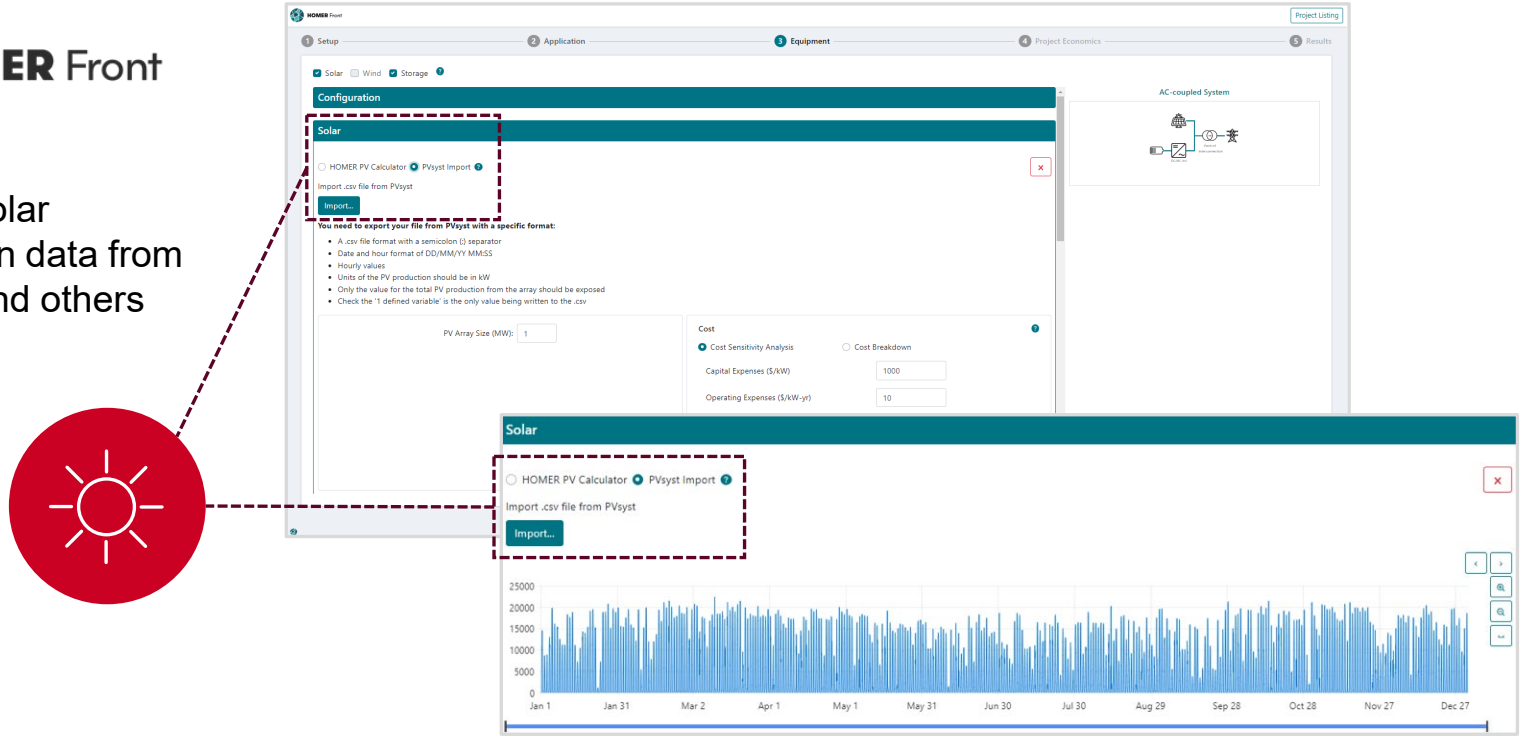


Software screen captures throughout are owned by UL Solutions

Model setup – renewable energy production data

HOMER Front

- **Import** solar production data from PVsyst and others

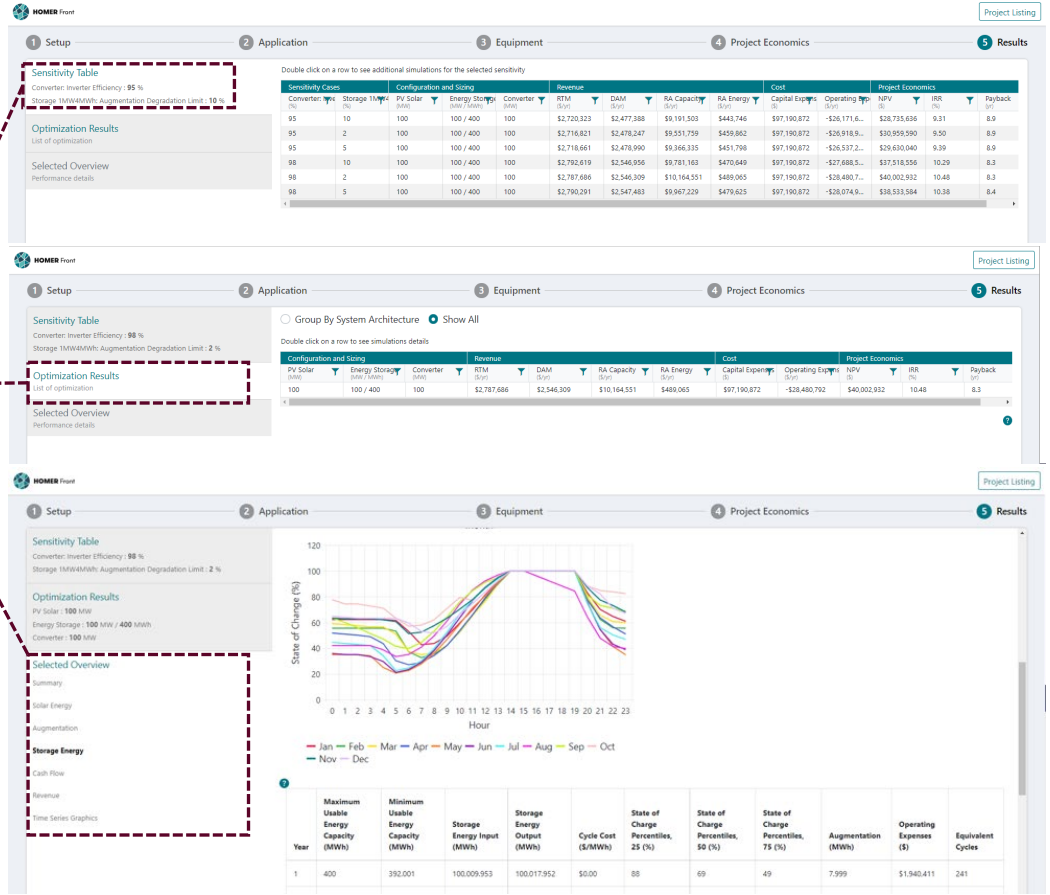


Software screen captures throughout are owned by UL Solutions

Results

HOMER Front

- **Sensitivity analysis**
  - Optimization results for specific sensitivity
  - Selected system detail on revenue, payback period, energy sold

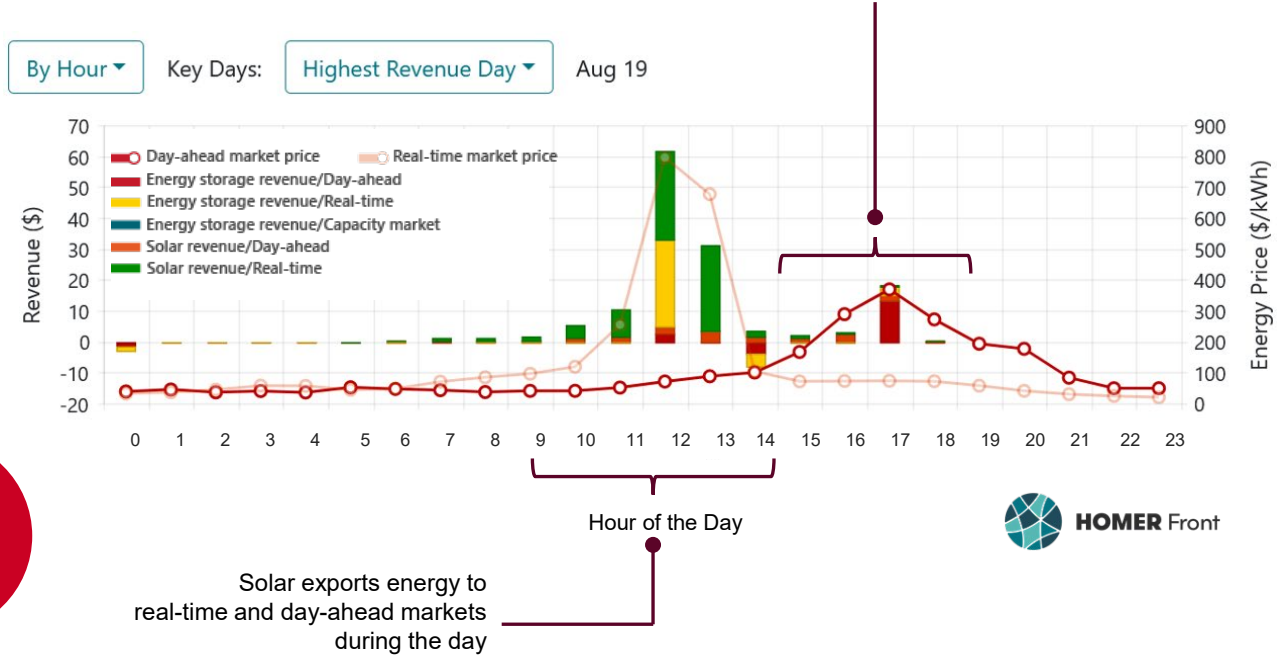


Software screen captures throughout are owned by UL Solutions

Results – revenue and price chart

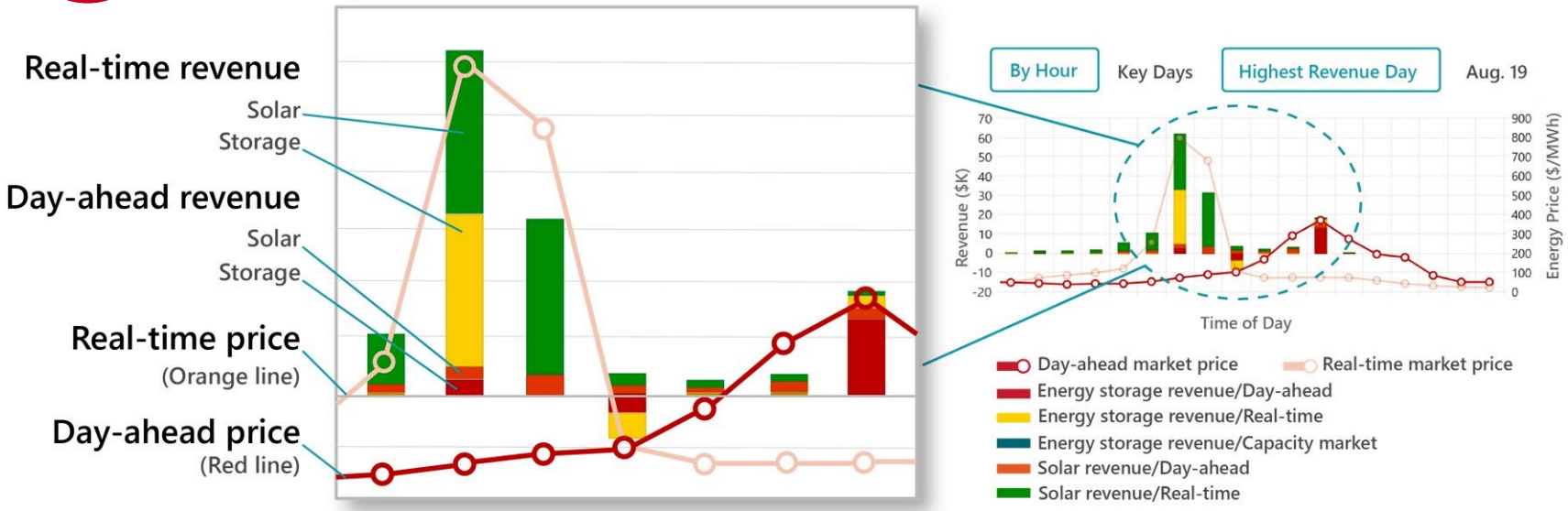
Selected system revenue over one day by equipment and market

- **Each bar = one hour**
  - Solar daytime export, storage export in the evening to maximize revenue
  - Shows price signal
  - Storage generates revenue by energy shifting arbitrage



Results – revenue analysis

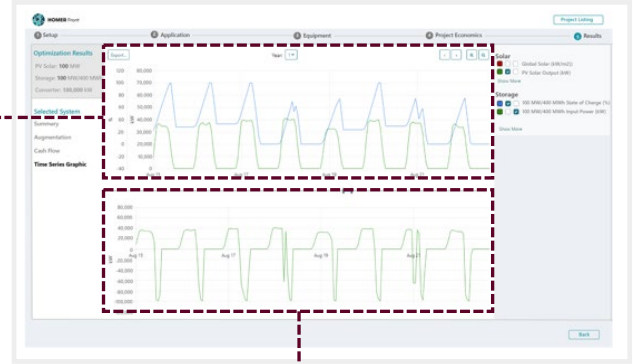
By discharging the battery when the price is high, battery energy storage system (BESS) helps to maximize revenue



Simulated system performance

Conduct sensitivity and risk analysis on key parameters such as equipment cost, locational marginal price (LMP) escalation, augmentation strategies







- **Graphics depict**
  - Energy storage charge and discharge activity
  - System performance
  - Sources of revenue

Software screen captures throughout are owned by UL Solutions

HOMER Front

Utility-scale front-of-the-meter

Case studies



ASP22CS460808

Determine financial impact of energy storage



UL Solutions helped California-based developer evaluate revenue

Client	**California-based developer**
Problem	• Developer needs to determine if adding energy storage to a solar plant will increase annual revenue and ROI and serve the load of the plant's utility offtaker. • The utility offtaker needs to meet expected electrical demands and greenhouse gas reduction and renewable energy commitments. • The project will participate in the California ISO day-ahead and real-time markets. • Developer must determine if additional revenues from energy storage will offset the added costs.
Solution	• Analysis of the 100 megawatts (MW) PV and Storage of 100 MW/400 Megawatt hours (Mwh) system yields the following results: • Initial equity investment – $104 million • 20-year internal rate of return – 7.8% • Payback – 13 years
HOMER Engagement	• UL Solutions conducted a feasibility study using HOMER Front modeling software to analyze the financial impact of solar-plus-storage. • The team modeled participation in the wholesale energy markets using hourly and sub-hourly locational marginal pricing from a node within a primary North-South transmission line in California. The model also included a 2020 resource adequacy (RA) network event schedule in California independent system operator (ISO) markets.



Project	Location	Technologies modeled	Revenue services
PV solar + energy storage	Northern California, U.S.	PV 100 MW Storage: 100 MW/400 MWh	Time of delivery Day-ahead Real time Resource adequacy

Payback – 13 years

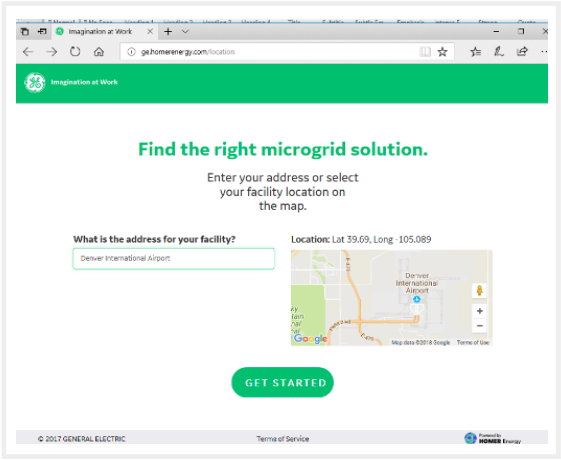
$	Year-one revenue
6.9M	Resource adequacy – capacity
2.4M	Resource Adequacy – day-ahead
2.4M	Day-ahead energy market
2.6M	Real-time energy market

Other services

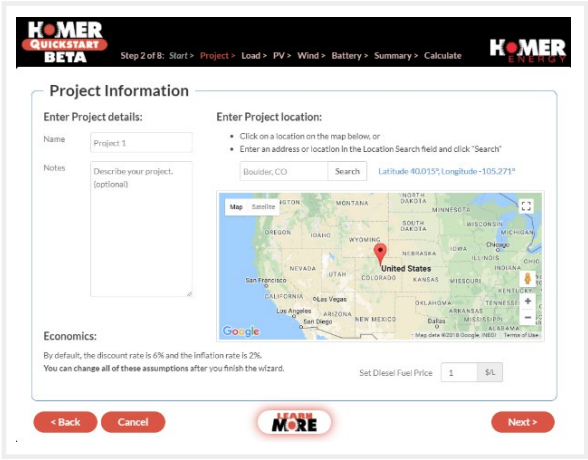


SaaS-API for HOMER Pro and HOMER Grid

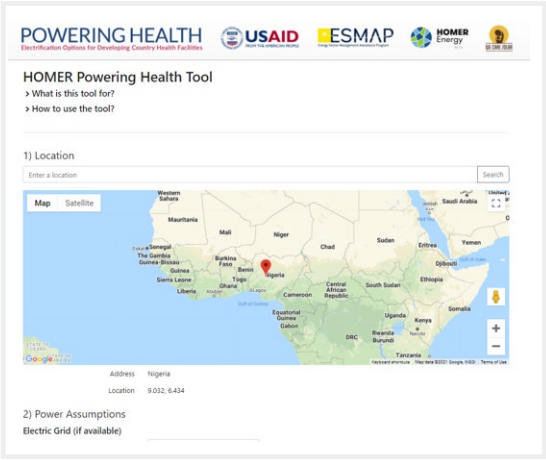
Software as a Service (SaaS) puts the power of HOMER in the cloud and allows the development of customized interfaces specific to your application.



Try the GE SaaS tool
http://ge.homerenergy.com



Try HOMER Quickstart
http://quickstart.homerenergy.com/



Try USAID Powering Health
http://tools.poweringhealth.org/

Training and support

Training

- Each HOMER license includes 2.5 hours of live training to get you started.
- Additional training classes are conducted monthly.
- Advanced training programs are available.

Support

- Manuals are continuously updated
- Wizard-based setup assistant
- Numerous sample files
- Quick and easy knowledge base
- Videos from software introduction to specific features available
- Engineering support is available within 24 business hours; commonly used for model verification and mini-trainings

www.homerenergy.com/support



Thank you

Try HOMER Pro and **HOMER Grid**

Complimentary 21-day trial

HOMER Pro
Remote, standalone microgrids
homerenergy.com/TryPro

HOMER Grid
Grid-connected DERs
homerenergy.com/TryGrid

HOMER Front
Utility-scale storage
with or without solar and wind
homerenergy.com/

UL.com/HOMERsoftware



UL Solutions

Thank you

UL.com/Solutions

UL.com/HOMERSoftware

© 2022 UL LLC. All rights reserved.